

Tradeoffs between carbon assimilation and release reveal a key role of the tropics in long-term global-scale carbon sequestration

Estefanía Muñoz^{a,b,1} and Carlos A. Sierra^b

This manuscript was compiled on March 6, 2025

The quantification of terrestrial ecosystems' carbon sequestration is crucial for understanding their role in mitigating global warming, with proper assessment requiring consideration of temporal scale. Our study quantified carbon sequestration by integrating carbon inputs and outputs over time to analyze the trade-off between high carbon assimilation rates associated with the tropics and low carbon losses through respiration related to high latitudes, while also assessing the effect of disturbance-related carbon losses. For the analyses, we utilized results from 9 CMIP6 Earth System Models. Then we compared the findings with the results of non-coupled models (TRENDY) and in-situ measured data-based products (CarboScope, FLUXCOM-X, CAMS). Results show that when only assimilation and respiration are considered, tropics exhibit higher carbon sequestration during 1850-2014, but the inclusion of disturbances shifts higher sequestration to high latitudes. Comparisons among products in 2000-2014 reveal similar trends between CMIP6 and FLUXCOM-X for carbon sequestration without disturbances, albeit with lower TRENDY magnitudes. When disturbances are considered, most products indicate sequestration amounts in high latitudes, except CAMS, which favors mid-latitudes. These findings underscore the significant impact of disturbances on terrestrial carbon cycles over the past century, suggesting that ecosystems, particularly in the tropics, had the potential to sequester more carbon than they did.

Global warming mitigation | Terrestrial ecosystems | CMIP6 | Carbon cycle

The capacity of the terrestrial biosphere to take up and store about a third of anthropogenic carbon emissions derived from the combustion of fossil fuels (1) depends to a large degree on the long-term balance between its carbon inputs (assimilation) and outputs (release), which fluctuate over time and space due to natural processes and human activities (2). Carbon inputs to terrestrial ecosystems are controlled primarily by photosynthesis, while carbon outputs arise from respiration of vegetation and soils as well as disturbances such as wildfires and land-use changes (3).

Although there is good evidence that the terrestrial biosphere has acted as a significant sink of carbon during the previous decades (1, 4), there is large uncertainty regarding the stability and persistence of this sink (5). For instance, measurements of atmospheric radiocarbon have shown that the terrestrial biosphere switched from a sink to a source of radiocarbon produced by thermonuclear-weapon testing about two decades after its spike in the atmosphere (6, 7). Similarly, recent measurements of radiocarbon in ecosystem respiration have shown a one- to two-decade time lag between photosynthesis and respiration (8, 9). These measurements indicate that carbon uptake from the terrestrial biosphere is temporary, and the large majority of carbon fixed by photosynthesis returns to the atmosphere in a matter of years to a few decades (10). Therefore, the large land carbon sink on land may soon be overtaken by a lagged respiratory response that could turn the land biosphere into a net carbon source.

A key metric to understand the temporary nature of the balance between carbon assimilation and release, as well as the lag times between carbon sinks and sources, is the transit time of carbon. This metric, however, is very difficult to determine for the nonlinear and non-equilibrium conditions of the coupled carbon-climate system (11). Recent studies have shown that, for long time periods, it is possible to obtain the cumulative lag effects between assimilation and release, implicitly defined by transit times, without explicitly computing this metric (12). Instead, an approach that relies on the mathematical integration (double integral) between instantaneous sources and sinks can be employed. This approach, which we define here as integrated Carbon Sequestration (CS), can be applied to the large volumes

Significance Statement

Understanding terrestrial ecosystems' role in carbon sequestration is crucial for addressing global warming. This study provides key insights by analyzing carbon dynamics across latitudes and time scales, incorporating both natural processes and human-induced disturbances. By comparing various models and data products, we reveal that tropical regions show higher carbon sequestration potential when considering only assimilation and respiration. However, including disturbances shifts this advantage to higher latitudes. These findings highlight the significant impact of disturbances on global carbon cycles over the past century and suggest untapped sequestration potential, particularly in tropical ecosystems. Our research emphasizes the importance of comprehensive, long-term assessments in developing effective climate mitigation strategies and underscores the need for targeted conservation efforts to maximize the carbon sequestration capacity of terrestrial ecosystems worldwide.

Author affiliations: ^aCREAF, Cerdanyola del Vallès, Barcelona, Catalonia, Spain; ^bTheoretical Ecosystem Ecology Group, Max Planck Institute for Biogeochemistry, Jena, Germany; ^cAffiliation Three

Please provide details of author contributions here.

Please declare any competing interests here.

¹Estefanía Muñoz. E-mail: e.munoz@creaf.uab.cat

of available information on net carbon fluxes between the land and atmosphere. The NOAA-GFDL-ESM4 and ACCESS-ESM1-5 models show the highest CS values, while the CanESM5 and BCC-CSM2-MR models show the lowest values (Fig. 2a). All models predict that tropical regions retained the largest amount of carbon from 1850 to 2014, followed by mid-latitudes (Fig. 2b-d). Although high latitudes sequester on average more carbon per unit area than mid-latitudes (Fig. 1b), the larger territorial expanse of mid-latitudes results in a greater total CS (Fig. 2). All CMIP6 models indicate a positive CS in the tropics during the studied period (Fig. 2b). In mid- and high-latitudes, at least one model (BCC-CSM2-MR) shows negative CS, where the accumulated balance between respiration and GPP results in a net loss over the 165 year period (Fig. 2c,d). Additionally, three models indicate nearly zero CS in high latitudes (Fig. 2d).

Our aim in obtaining this metric from existing information is to learn from the historical record on the trade-off between carbon sinks and lagged carbon release. For example, although tropical forests have large photosynthetic fluxes, carbon returns back to the atmosphere relatively quickly (9, 13). On the contrary, high-latitude ecosystems with lower photosynthetic rates return carbon to the atmosphere more slowly than tropical ecosystems (14, 15). Therefore, it is uncertain if the capacity of the terrestrial biosphere to store carbon is mainly controlled by large photosynthetic inputs or slow lagged respiratory fluxes.

We calculated CS for the period 1850–2014 using monthly outputs from gross primary productivity (GPP), heterotrophic respiration (r_h), autotrophic respiration (r_a), and net biome production (NBP), based on output from nine Earth System Models (ESMs) participating in the Coupled Model Intercomparison Project Phase 6 (CMIP6). We also used numerical output from land surface models participating in the TRENDY project. To contrast modeling results with observations, we calculated CS for a shorter period (2001–2014) using results from two global atmospheric inversions that assimilate data on mole fraction of CO_2 and provide gridded net exchange fluxes of carbon between the land and the atmosphere (CarboScope, and CAMS), and a product that assimilates data on net ecosystem exchange, land surface cover, and meteorology (FLUXCOM-X).*

By accurately quantifying terrestrial carbon sequestration, we aim to deepen our understanding of ecosystem carbon dynamics to inform better land management practices and enhance strategies to maximize ecosystems' capacity to mitigate climate change effectively.

Results

Spatial patterns of Carbon Sequestration integrated over 165 years. Carbon Sequestration (CS) obtained as the double integral of NEP from eight CMIP6 models (Supplementary Table 1 (see SI.tex)) reveal a predominantly positive global distribution, with notable exceptions in polar regions and hyper-arid zones (Fig. 1a). Forests exhibit the highest CS values, particularly in the Amazon, Eastern and Western North America, the Congo basin, Western Europe, and South East Asia. The latitudinal CS average shown in Fig. 1b demonstrates positive CS across all latitudes, indicating long-term carbon retention globally during the entire 165 year simulation period (1850–2014). Peak CS occurs in tropical latitudes (approximately 25°S to 20°N), while mid-latitudes (approximately $20\text{--}40^\circ\text{N}$ and $20\text{--}30^\circ\text{S}$) show the lower values.[†] Higher values of CS in tropical and boreal regions in comparison to mid-latitudes suggests that high C assimilation or low C release contribute to retain carbon in the long-term, but the effect of high C assimilation is stronger as suggested by the high CS values in the tropics.

Integrated CS across all latitudinal zones exhibits a consistent upward trend over time in all models, albeit with

Disturbance impacts on Carbon Sequestration integrated over 165 years. The previous results, based on the double integral of NEP, only account for differences between GPP and ecosystem respiration and exclude fluxes due to disturbances like deforestation and fires. When using Net Biome Production (NBP) to calculate CS, spatial and temporal dynamics change significantly, with values about two orders of magnitude smaller than those obtained with NEP (Figs. 2a and 3a). These values are predominantly negative, which indicates that over the entire 165-year timeframe the accumulated carbon losses are higher than carbon gains producing a net carbon debt. Globally accumulated CS is negative for all CMIP6 models except CNRM-ESM2-1 and ACCESS-ESM1-5, with the ensemble indicating a carbon debt of approximately 0.8 PgC yr for the period 1850–2014 (Fig. 3a). High-latitude regions show the least negative values, followed by the tropics (Fig. 3b-d). Negative CS values are observed in 5 out of 8 models for the tropics, 6 for mid-latitudes, and 5 for high latitudes (Fig. 3b-d). Notably, the ensemble accumulation of CS over time is no longer monotonically increasing as when using NEP; after decreasing from 1850 to around 1970, CS tends to slightly increase showing a recovery trajectory.

Comparison with observation-based products. For the 14-year period when model output and observation-based products overlap (2001–2014), integrated CS consistently show higher values when derived from NEP compared to NBP (left and right panel of Fig. 4, respectively). NEP-based CS from CMIP6, TRENDY, and FLUXCOM-X indicate that tropical regions have the highest values, while high latitudes have the lowest, aligning with previous results in Fig. 2. CMIP6 ensemble yields the highest global CS values, while TRENDY shows the lowest. Although both ensemble models display similar latitudinal zone trends, CMIP6 consistently surpasses TRENDY by up to 250 PgC yr^{-1} globally. This substantial difference observed despite some TRENDY models being used in CMIP6 Earth System Models underscores the impact of coupling terrestrial models with climate models and the explicit modeling of carbon movement throughout the Earth System in CMIP6. Then, when using NBP, mid-latitudes show the highest CS values for CMIP6, TRENDY, and CAMS, but not for CarboScope (left panel in Fig. 4). This finding is consistent with the CMIP6 results for the 1850–2014 period. However, when examining the common 14-year timeframe (2001–2014) across all products, CS values are predominantly positive, except for CAMS in tropical regions. This shift

*I think it's better to provide the citations of these products than the links. The links won't work in a few years, and the authors usually prefer a citation.

[†]Figure 1 can be improved by moving the legend to the right so it doesn't overlap with the the results for low latitudes. The labels a and b do not need a parenthesis, you can remove them.

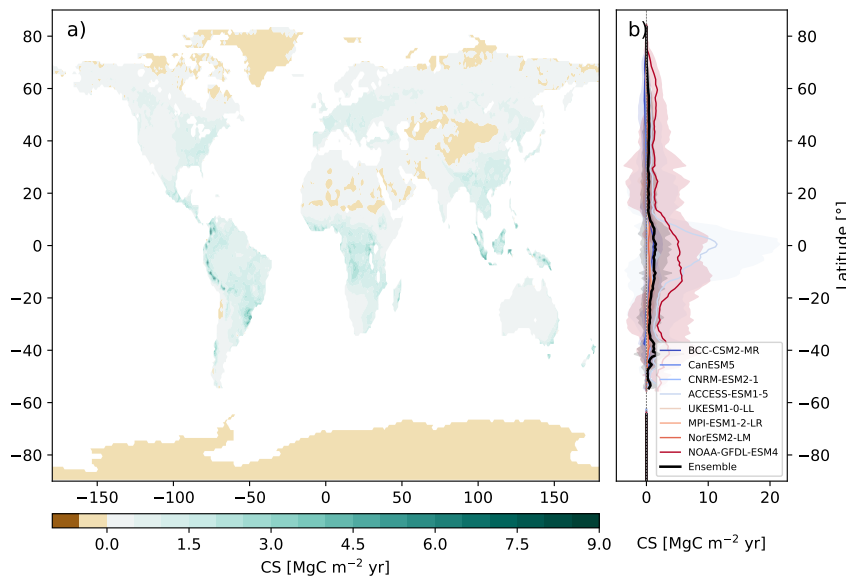


Fig. 1. Carbon Sequestration (CS) spatial patterns from NEP from CMIP6 models (1850-2014). a) Global distributions of CS based on the weighted ensemble of 8 CMIP6 models, and b) latitudinal CS profiles for individual CMIP6 models (colored lines) and their ensemble (black line). The dotted line in b) represents null values of CS and the shadow areas indicate two standard deviations from the mean.

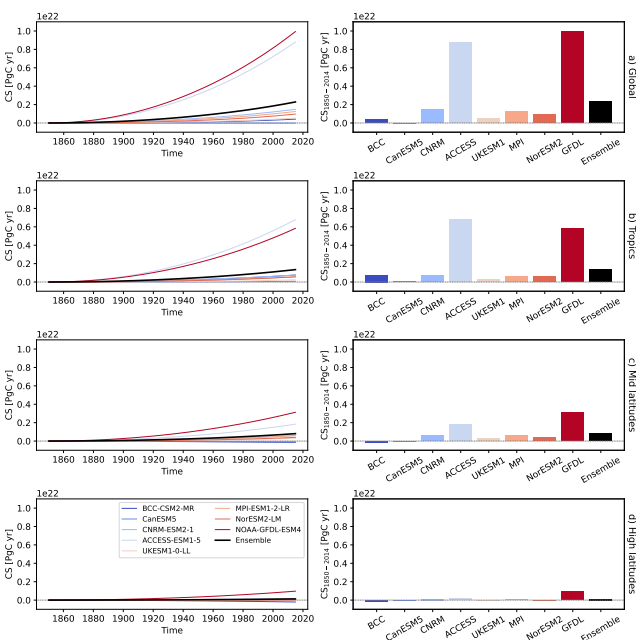


Fig. 2. Time accumulation of Carbon Sequestration (CS) (left panel) and CS at the end of the period from 1850 to 2014 (right panel) calculated using NEP for a) global, b) tropics, c) mid-latitudes, and d) high-latitudes zones calculated using the results of 8 models from CMIP6. Black lines and bars indicate the results using the ensemble of the models. The dotted lines indicate null values of CS. To save space, only the initial part of each model name is displayed on the horizontal axes in the right panel.

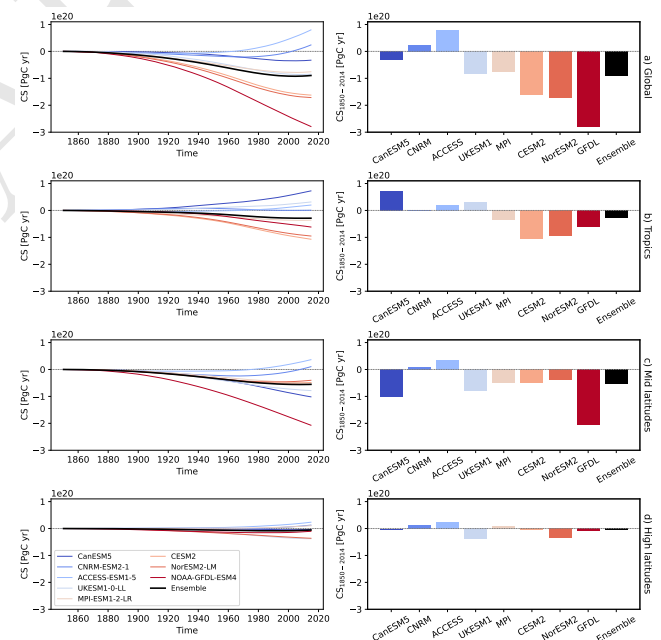


Fig. 3. Time accumulation of Carbon Sequestration (CS) (left panel) and CS at the end of the period from 1850 to 2014 (right panel) calculated using NBP for a) global, b) tropics, c) mid-latitudes, and d) high-latitudes zones calculated using the results of 8 models from CMIP6. Black lines and bars indicate the results using the ensemble of the models. The dotted lines indicate null values of CS. To save space, only the initial part of each model name is displayed on the horizontal axes in the right panel.

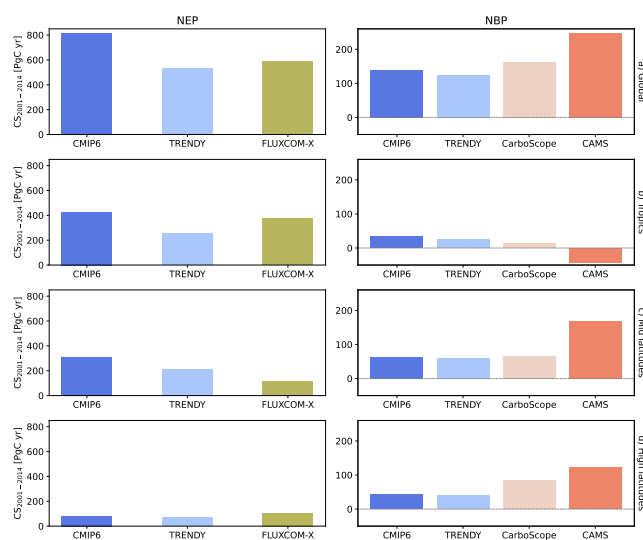


Fig. 4. Carbon sequestration (CS) across different products (2001-2014) calculated using NEP or NEE (left panel) and NBP (right panel) for a) global, b) tropics, c) mid-latitudes, and d) high-latitudes. The dotted lines indicate null values of CS. CMIP6 and TRENDY CS values correspond to the weighed ensemble of their respective models. Note that the y-axis of the left and right panels do not have the same scale.

towards positive values likely stems from the increasing trend in CS observed in Fig. 3 from around 1970 onwards. In this shorter timeframe, tropical regions shift from having the second-highest CS to the lowest, with CAMS even indicating a negative value. For NBP-based calculations, inversion models (CAMS and CarboScope) show higher CS values than the model inter-comparison products. CMIP6 and TRENDY yield similar CS values when using NBP, with CMIP6 slightly higher, but the difference is less pronounced compared to NEP-based calculations.

Discussion

Trade-off between high carbon assimilation and slow lagged respiration. Terrestrial ecosystems worldwide have absorbed more carbon than released from 1850 to 2014, acting as net carbon sinks when disturbances are not considered. This is evident from the positive CS values across all latitudes (Fig. 1b) and the steady increase in the left panel in Fig. 2. These findings align with the growing global carbon budget and previous research that has observed increases in NEP over various periods and using different methods and databases (1, 4, 16, 17). Several factors explain this trend, including increased atmospheric CO₂, higher nitrogen deposition, and the effects of climate change. These factors enhance photosynthesis via fertilization, improve plant water use through reductions in stomatal conductance in water-limited ecosystems, reduce cold limitations at high latitudes, and extend growing seasons (1, 16, 18–20). While elevated CO₂ can positively and negatively affect carbon sequestration, the overall impact up to 2014 appears positive (right panel in Fig. 2). The trends remain consistent even when focusing solely on the 2001-2014 period (Left panel in Fig. 4).

The high CS observed in forests during the studied period (Fig. 1a) supports earlier research indicating that forests, particularly in tropical and boreal regions, are the primary

reservoirs of the world's biomass carbon (21). This is likely because forests are dominated by trees with large woody masses, and most of the carbon captured by vegetation is allocated to woody tissues (22). Their complex structure also makes forests highly efficient at capturing carbon, and when vegetation dies, it contributes to soil organic carbon, further enhancing storage (23). Over time, increases in NEP have been observed in forests, especially in those without nutrient limitations (16, 18, 24).

The tropics show higher carbon sequestration values when data are divided by latitudinal zones (Figs. 1b and 2b), indicating that high tropical assimilation outweighs low respiration in extratropical regions when only climate is considered. This can be attributed to the dense vegetation, extensive biomass, and rapid growth in tropical areas, which contribute significantly to NEP trends (1, 16, 21, 25, 26), even with rising temperatures and lower simulated soil carbon stocks and residence times (27, 28). High and stable temperatures in these regions promote increased carbon allocation in the canopy (22). However, it's important to note that CMIP6 models do not capture potential declines in tropical forest carbon sink strength due to carbon saturation and drought, as suggested by several studies (20, 26, 29). In contrast, high latitudes (>50°) exhibit lower CS values, likely due to temperature limitations, although they show enhanced seasonal CO₂ cycles with larger summer uptakes and winter releases (30). Despite being well-studied, carbon sinks in high-latitude regions still show discrepancies among different products, and soil carbon is often underestimated, suggesting a potentially greater carbon sequestration capacity than currently simulated (20, 31).

Consequences of disturbances. When factors beyond climate, such as harvesting, deforestation, and fire, are considered, the carbon sequestered is significantly lower. From 1850 to 2014, most models indicate that cumulative respiration exceeded assimilation (Fig. 3), making terrestrial ecosystems a carbon source rather than a sink. While the Amazon, Congo, and Papua New Guinea forests still exhibit the highest CS values (Supplementary Fig. 5), high-latitude regions show the smallest carbon losses (Fig. 3). The change in tropical dynamics is driven by deforestation, degradation, and drought, especially in the Amazon, as well as fires and the negative effects of climate variability and change (20, 21, 26, 29, 32). Although African tropical forests generally show net carbon gain and intact tropical Americas continue to gain carbon (20, 21, 26) (though the sink is declining (33)), cumulative carbon fluxes over the past 165 years before 2014 indicate that ecosystems in tropical latitudes have experienced a net loss of carbon. Less negative CS values at high latitudes can be attributed to lower respiration rates. Additionally, northern boreal forests seem to be benefiting from climate change and accumulating carbon at higher rates than in the past (1, 21, 26, 34).

When only data from 2001-2014 is considered, CS calculated with NBP becomes positive, albeit much lower than CS calculated with NEP for the same period (Fig. 4). This shift in trend demonstrates how carbon flux balances can vary significantly depending on the chosen time horizon. It shows that over these 14 years, carbon inputs can offset recent losses. However, these short-term gains are insufficient to counteract the long-term accumulated carbon debts. These gains could

also be noted in the change of slope observed from around the 1970s for based-NBP CS (left panel in Fig. 3a). Several factors could explain the carbon gains observed from 2001-2014 and the positive trend since the 1970s. These include increased CO₂ fertilization, longer growing seasons, enhanced nutrient cycling and nitrogen deposition, and reforestation efforts (20, 35). Additionally, the forest transition in the 19th and 20th centuries, followed by forest regrowth in Europe (20), land abandonment, preservation efforts, improved land management practices during the second half of the 20th century (36), and the acceleration in global temperature since the 1970s (26) could all have contributed to this trend.

The tropics sequestered the least amount of carbon among latitude zones during 2001-2014 (right panel in Fig. 4), with CAMS even indicating a net carbon release. This aligns with previous findings indicating high cumulative land-use emissions in the tropics and that warm climates decrease NBP due to temperature and soil moisture sensitivities (20, 34, 37, 38). It's also important to acknowledge that while CMIP6 models capture carbon gains from tree growth, they don't accurately represent carbon losses from tree mortality (29).

Agreement and discrepancies among products. Carbon sequestration varies significantly among CMIP6 models and different products as showed before using different metrics (17, 31, 34, 39), influenced by factors such as CO₂ response, temperature perturbations, resolutions, residence times, land surface schemes, and the inclusion of components such as nutrient cycles, fires, and dynamic vegetation (27, 34, 40, 41). Models coupling the nitrogen (N) cycle are expected to reduce CO₂ fertilization and carbon releases, and align better with observations (41, 42).

Our results differ from previous CMIP6 analyses. While Arora et al. (34) and Melnikova et al. (39) identified CNRM-ESM2-1 and CanESM5 as having the largest land carbon uptake, our results only show CNRM-ESM2-1 and only as the second largest CS when disturbances are considered. These models, which lack nitrogen cycle coupling, do not demonstrate notably high CS values. Interestingly, our study shows ACCESS-ESM1-5 as one of the models with the highest CS values for both NEP and NBP, being among the few indicating positive NBP-based CS (Figs. 2 and 3). This contrasts with Arora et al. (34)'s findings, which reported this model as having the lowest land carbon uptake due to nutrient limitations from N and P coupling.

The discrepancies in our findings likely arise from the carbon sequestration metric used in this study. Unlike approaches focusing solely on carbon stocks, CS considers the long-term balance between carbon inputs and outputs, accumulating effects over time (43, 44). A possible explanation for the high CS values in ACCESS-ESM1-5 could be that the model captures both the low carbon uptake rates caused by N limitations and the small carbon releases due to additional nitrogen release from decomposing organic matter triggered by rising global temperatures (41, 45), and thus, CS balance both fluxes over time.

Our analysis revealed no clear pattern between models considering N cycle coupling (ACCESS-ESM1-5, CESM2, MPI-ESM1-2-LR, NorESM2-LM, UKESM1-0-L) or dynamic vegetation (NOAA-GFDL-ESM4, CESM2, and UKESM1-0-LL) and CS. However, models incorporating fire components

(NOAA-GFDL-ESM4, NorESM2-LM, CESM2, MPI-ESM1-2-LR) generally showed lower CS-NBP values (Fig. 3). Notably, the NOAA-GFDL-ESM4 model shifted from having the highest CS value for NEP to the lowest for NBP (Figs. 2 and 3), suggesting that immediate carbon losses from fires outweigh the potential long-term benefits of low-intensity fires in promoting forest carbon storage (46), or that these effects are not considered in the models.

Comparing 2001-2014, results from CMIP6, TRENDY, and FLUXCOM-X agree that CS without disturbances is highest in the tropics, followed by mid-latitudes (left panel in Fig. 4). This pattern aligns with the full CMIP6 time series (Fig. 2). CMIP6 ensemble shows the highest global CS values when calculated with NEP, while the TRENDY ensemble shows the lowest as noted for carbon land sink (39). For CS-NBP, CMIP6 remains higher than TRENDY, but the gap narrows (Fig. 4). The main difference between CMIP6 and TRENDY lies in their nature: CMIP6 is an intercomparison of ESMs, while TRENDY of DGVMs. ESMs assimilate physical atmospheric and oceanic data, reproducing historical CO₂ growth rates and carbon fluxes between atmosphere, sea, and land (20, 47). Note that these products are not entirely independent, as some CMIP6 models use TRENDY as land models.

TRENDY and CMIP6 models differ from CAMS and CarboScope inversion products, with the latter showing higher global carbon sequestration (CS-NBP) except in the tropics, where CAMS even indicates negative CS values (right panel in Fig. 4). These discrepancies may arise because intercomparison models often inadequately parameterize land management and degradation (16), underestimate carbon ecosystem residence times (48), fail to fully capture the impact of volcanic eruptions, (49), underestimate soil carbon stocks in northern latitudes (31), and poorly reproduce forest carbon stocks (40) potentially due to limitations in modeling recruitment, mortality, and vegetation composition changes, as well as climatic and CO₂ interactions (16). In the tropics, CMIP6 and TRENDY show higher CS-NBP values than inversion products (Fig. 4b), likely due to modeling shortcomings. These include misbalancing CO₂ fertilization and respiration costs (29), inaccurate mortality trends (29), overestimating biomass (50), and underestimating drought impacts in tropical forests (26). Additionally, models tend to overestimate carbon in tropical regions while underestimating it in the Northern Hemisphere (42). It should be noted that inversion models also have inherent uncertainties due to data limitations, design choices, and under-sampling in certain biomes, as in the tropics (1). These uncertainties, along with variations in interhemispheric transport models and inversion settings (16), lead to differing results between CarboScope and CAMS, although the specific causes for these differences are not always clear (38). While this study didn't directly compare FLUXCOM-X with inversion models, previous research found agreements between FLUXCOM-X and CarboScope for 2015-2020 (25). Here, FLUXCOM-X shows higher CS values for high latitudes compared to TRENDY and CMIP6, possibly due to its incorporation of more observational data, which may better capture complex processes in northern extra-tropics (20).

Implications for climate change mitigation. Despite renewable energy growth and some decarbonization, fossil fuels still

dominate to meet rising global energy demand, causing emissions to increase and hindering Paris Agreement goals (20, 51). While terrestrial ecosystems help mitigate these emissions (18, 20, 52), some regions are carbon neutral or sources, and future projections suggest ecosystems may become net carbon sources overall (20, 39). Our results show that terrestrial ecosystems acted as carbon sinks across all latitudes when considering only climate variations over the 165 years leading up to 2014 (Fig. 1). However, disturbances can shift this balance negatively (Fig. 3a). Interestingly, even when considering disturbances, CS was positive globally from 2001–2014, except in the tropics according to CAMS (right panel in Fig. 4). These findings underscore how the chosen time horizon can significantly influence whether an ecosystem appears as a carbon source or sink. This could be explained by the various processes operating on different time scales, which may not be fully captured in short-term carbon flux measurements (53, 54). Therefore, when evaluating carbon dioxide removal, capture, and storage strategies, it's crucial to quantify carbon sequestration by considering not only how much carbon is in the ecosystem, but how long it remains there (43, 55).

Past disturbances such as deforestation, fires, and land-use changes have significantly impacted carbon sequestration by terrestrial ecosystems, particularly in the tropics. The substantial difference between CS calculated with NEP and NBP underscores the high potential of terrestrial ecosystems to sequester carbon. This is particularly noted in the tropics, emphasizing the urgent need for preservation and sustainable management of tropical ecosystems as a priority in anthropogenic CO₂ mitigation.

While this work provides clear insights, significant discrepancies exist between the products used, particularly in the tropics and high latitudes (20). Earth System Models and Dynamic Global Vegetation Models are crucial for projecting future climate scenarios (1) but still struggle to fully capture complex natural interactions (1, 16, 37, 56). This highlights the need for continuous improvement in model development and refinement, particularly regarding carbon ecosystem residence time (27, 31, 34, 48, 53). Considering that multi-model ensembles of intercomparison products have been found to outperform individual models (40, 57), the weighted ensemble calculated for TRENDY and CMIP6 in this study could potentially reduce model bias.

In conclusion, this work emphasizes the importance of considering time when assessing carbon sequestration. Furthermore, the spatial patterns of carbon sequestration, both with and without disturbances, can inform the prioritization of areas for targeted climate change mitigation efforts through carbon sequestration strategies.

Materials and Methods

Datasets. We calculated the carbon sequestration using Net Ecosystem Production (NEP) (or Net Ecosystem Exchange (NEE)) and Net Biome Production (NBP) outputs from two model intercomparison products and three data-driven products.

Model intercomparison products. We used the results of Gross Primary Production (GPP), autotrophic respiration (r_a), heterotrophic respiration (r_h), and NBP from a set of Earth System Models (ESMs) participating in the most recent climate-carbon simulations under the sixth phase of the Coupled Model Intercomparison Project (CMIP6) (58). Eight models provided monthly outputs for NBP, while eight (seven overlapping) supplied data for the remaining variables of interest in the esm-hist simulation (see Supplementary Table S1). These simulations span the period from 1850 to 2014 and are forced by prescribed anthropogenic CO₂ emissions rather than CO₂ concentrations. The data were obtained from The German Climate Computing Centre DKRZ. In addition, we used results from Dynamic Global Vegetation Models (DGVMs) compiled by the TRENDY v13 project (26). Twenty models provided monthly outputs of GPP, r_a , r_h , and 18 models provided monthly NBP (see Supplementary Table S2). We used experiment S3, which accounts for variations in atmospheric CO₂ and climate and spans the period from 1700 to 2023. Data were requested through the Global Carbon Budget.

Both CMIP6 and TRENDY intercomparison models were included in the analysis due to their distinct focuses. TRENDY DGVMs focus on simulating terrestrial vegetation dynamics and ecosystem processes, while CMIP6 ESMs provide a representation of the entire Earth system, incorporating simplified versions of DGVMs as subcomponents within a broader framework that also accounts for atmospheric, oceanic, and cryospheric processes (59, 60). For both CMIP6 and TRENDY models, NEP values are calculated as $NEP = GPP - (r_a + r_h)$.

Data-driven products. Besides the model intercomparison products, we utilized data from two atmospheric CO₂ inversion products: CarboScope and CAMS. These products estimate land-atmosphere net carbon exchange fluxes by combining CO₂ concentration measurements, transport models, and information on anthropogenic carbon emission and land-atmosphere carbon exchange (61). Specifically, we used the nbetEXToc_v2024E version of CarboScope (62), which provides daily data from 1957 to 2023 based on measurements from 173 sites. For CAMS (63), we used version v24, which includes monthly data from 1979 to 2020 derived from over 100 surface measurements sites. Additionally, we incorporate Net Ecosystem Exchange (NEE) from X-BASE produced with FLUXCOM-X (25), a machine-learning-based product trained on in-situ eddy covariance data. FLUXCOM-X provides monthly NEE estimates for the period 2001–2021.

In this study, we assume that NEE and NEP are interchangeable since both represent the same underlying processes—GPP minus respiration—despite being calculated using different data sources. Note the NEE is negative when the ecosystem assimilates more CO₂ than it releases, which is opposite to the convention for NEP.

Homogenization and ensemble calculations. To enable comparison, data from all products were regridded to a uniform 1° × 1° spatial resolution using the bilinear interpolation method implemented via xESMF (64). We standardized the temporal resolution to monthly for all datasets that were not originally in monthly format. CMIP6 calculations utilized the full-time series (1850–2014), while comparisons across products were restricted to their overlapping period, i.e., 2001–2014. Positive NEP and NBP values were defined as fluxes moving from the atmosphere to the land, which contrasts with the conventions used in inverse models.

Then, to calculate the ensemble of CMIP6 and TRENDY models, we used a weighted averaging approach rather than a simple equal-weight average. This procedure assigns weights to each model based on its predictive performance against available data, following the multimodel inference framework presented by Burnham & Anderson (65), which builds on concepts developed by H. Akaike in the 1970s. The weighting procedure calculates each model's weight by measuring its distance from observed data.

However, since we are dealing with global spatial models where direct observations are not available, we used data-driven products derived from sparse observations and scaled to the terrestrial surface. Specifically, we used FLUXCOM-X to calculate the NEP ensembles and CarboScope for NBP ensembles (chosen for its longer time series compared to CAMS). The calculation of weights (w_i) for each model i is detailed in (cite text with method). Once w_i are determined, the model ensemble ($\bar{x}$) is computed as

$$\bar{x} = \sum_{i=1}^k w_i \cdot x_i \quad [1]$$

where k is the total number of models, and x_i are the values of the variable of interest for model i . In this case, x corresponds to NEP or NBP. The same weighting procedure is applied to create the ensemble of models from CMIP6 and TRENDY, using all data from periods that overlap with the available data periods of CarboScope (CMIP6:1957-2014, TRENDY:1957-2023) or FLUXCOM-X (CMIP6:2001-2014, TRENDY:2001-2021) (see Supplementary Figs. 1-4).

Carbon sequestration. After harmonizing all NEP (or NEE) and NBP data to the same temporal and spatial scale and calculating the ensembles for CMIP6 and TRENDY, integrated Carbon Sequestration (CS) is computed. Here, CS is defined as the storage of a certain mass of carbon (M) within a system over a given period ($t_0 + T$). It is quantified as the area under the curve representing the evolution of carbon mass over time (43), as

$$CS(t_0, T) := \int_{t_0}^{t_0+T} \int_{t_0}^t s(t) - r(t) dt dt. \quad [2]$$

where $s(t)$ is the gross flux rate of carbon removed from the atmosphere and stored into a system at a given time t , while $r(t)$ is the gross flux rate of carbon released from the system back to the atmosphere. This definition of CS accounts for both the fate of new carbon inputs and the legacy carbon already present in the system at t_0 (43). It considers the total carbon mass under evaluation as the net balance between all inputs and outputs over the specified time horizon (see Fig. 1 in Muñoz et al. (55)).

When calculating global carbon sequestration or aggregating it by latitudinal zones, the grid areas are factored into the computation, removing spatial units. Latitudinal zones are defined as tropical: 22.5°S-22.5°N, mid-latitudes: 22.5°S-55.0°S and 22.5°N-55.0°N, and high latitudes: >55.0°N.

ACKNOWLEDGMENTS. This project has received funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement No 101110350 (DISEQ - The Drought Impact on the Climate Benefit of Carbon Sequestration).

References.

1. S Ruehr, et al., Evidence and attribution of the enhanced land carbon sink. *Nat. Rev. Earth Environ.* 4, 518–534 (2023).
2. Y Malhi, DD Baldocchi, PG Jarvis, The carbon balance of tropical, temperate and boreal forests. *Plant, Cell & Environ.* 22, 715–740 (1999).
3. J Wisniewski, RK Dixon, JD Kinsman, RN Sampson, AE Lugo, Carbon dioxide sequestration in terrestrial ecosystems. *Clim. Res.* 3, 1–5 (1993).
4. AP Walker, et al., Integrating the evidence for a terrestrial carbon sink caused by increasing atmospheric CO_2 . *New Phytol.* 229, 2413–2445 (2021).
5. JG Canadell, et al., Saturation of the terrestrial carbon sink, The IGBP Series, eds. J Canadell, D Pataki, L Pitelka. (Springer-Verlag, Berlin-Heidelberg), pp. 59–78 (2007).
6. T Naegler, I Levin, Observation-based global biospheric excess radiocarbon inventory 1963–2005. *J. Geophys. Res. Atmospheres* 114 (2009).
7. I Levin, et al., Radiocarbon in global tropospheric carbon dioxide. *Radiocarbon* 64, 781–791 (2022).
8. A Tangarife-Escobar, et al., Radiocarbon isotopic disequilibrium shows little incorporation of new carbon in mineral soils of a boreal forest ecosystem. *J. Geophys. Res. Biogeosciences* 129, e2024JG008191 (2024).

9. I Chanca, et al., How long does carbon stay in a near-pristine central Amazon forest? An empirical estimate with radiocarbon. *Biogeosciences* 22, 455–472 (2025).
10. CA Sierra, GR Quetin, H Metzler, M Müller, A decrease in the age of respired carbon from the terrestrial biosphere and increase in the asymmetry of its distribution. *Philos. Transactions Royal Soc. A: Math. Phys. Eng. Sci.* 381, 20220200 (2023).
11. H Metzler, M Müller, CA Sierra, Transit-time and age distributions for nonlinear time-dependent compartmental systems. *Proc. Natl. Acad. Sci.* 115, 1150–1155 (2018).
12. H Metzler, S Launiainen, G Vico, Amount of carbon fixed, transit time and fate of harvested wood products define the climate change mitigation potential of boreal forest management—a model analysis. *Ecol. Model.* 491, 110694 (2024).
13. CA Sierra, LM Estupinan-Suarez, I Chanca, The fate and transit time of carbon in a tropical forest. *J. Ecol.* 109, 2845–2855 (2021).
14. N Carvalhais, et al., Global covariation of carbon turnover times with climate in terrestrial ecosystems. *Nature* 514, 213 (2014).
15. X Lu, YP Wang, Y Luo, L Jiang, Ecosystem carbon transit versus turnover times in response to climate warming and rising atmospheric CO_2 concentration. *Biogeosciences* 15, 6559–6572 (2018).
16. M Fernández-Martínez, et al., Global trends in carbon sinks and their relationships with CO_2 and temperature. *Nat. Clim. Chang.* 9, 73–79 (2019).
17. J Zhu, X Zeng, X Gao, H Zhang, Response of Terrestrial Net Primary Production to Quadrupled CO_2 Forcing: A Comparison between the CAS-ESM2 and CMIP6 Models. *Biology* 11, 1693 (2022).
18. M Fernández-Martínez, et al., Atmospheric deposition, CO_2 , and change in the land carbon sink. *Sci. Reports* 7, 1–13 (2017).
19. M O'Sullivan, et al., Have synergies between nitrogen deposition and atmospheric CO_2 driven the recent enhancement of the terrestrial carbon sink? *Glob. Biogeochem. Cycles* 33, 163–180 (2019).
20. P Friedlingstein, et al., Global Carbon Budget 2023. *Earth Syst. Sci. Data* 13, 1791–1805 (2023).
21. L Xu, et al., Changes in global terrestrial live biomass over the 21st century. *Sci. Adv.* 7 (2021).
22. X Lu, C Chung, Y Malhi, C Huang, Global Patterns in the Allocation of Forest Net Primary Production. *J. Biogeogr.* pp. 1–12 (2025).
23. Z Sha, et al., The global carbon sink potential of terrestrial vegetation can be increased substantially by optimal land management. *Commun. Earth Environ.* 3, 1–10 (2022).
24. S Stith, et al., Recent trends and drivers of regional sources and sinks of carbon dioxide. *Biogeosciences* 12, 653–679 (2015).
25. JA Nelson, et al., X-BASE: the first terrestrial carbon and water flux products from an extended data-driven scaling framework, FLUXCOM-X. *Biogeosciences* 21, 5079–5115 (2024).
26. S Stith, et al., Trends and Drivers of Terrestrial Sources and Sinks of Carbon Dioxide: An Overview of the TRENDY Project. *Glob. Biogeochem. Cycles* 38, 1–25 (2024).
27. A Ito, et al., Soil carbon sequestration simulated in CMIP6-LUMIP models: Implications for climatic mitigation. *Environ. Res. Lett.* 15 (2019).
28. L Zhang, et al., Mapping global distributions, environmental controls, and uncertainties of apparent top- and subsoil organic carbon turnover times (2025).
29. A Koch, W Hubau, SL Lewis, Earth System Models Are Not Capturing Present-Day Tropical Forest Carbon Dynamics. *Earth's Futur.* 9, 1–19 (2021).
30. R Commmane, et al., Carbon dioxide sources from Alaska driven by increasing early winter respiration from Arctic tundra. *Proc. Natl. Acad. Sci. United States Am.* 114, 5361–5366 (2017).
31. RM Varney, SE Chadburn, EJ Burke, PM Cox, Evaluation of soil carbon simulation in CMIP6 Earth system models. *Biogeosciences* 19, 4671–4704 (2022).
32. T Tagesson, et al., A physiology-based Earth observation model indicates stagnation in the global gross primary production during recent decades. *Glob. Chang. Biol.* 27, 836–854 (2021).
33. W Hubau, et al., Asynchronous carbon sink saturation in African and Amazonian tropical forests. *Nature* 579, 80–87 (2020).
34. VK Arora, et al., Carbon-concentration and carbon-climate feedbacks in CMIP6 models and their comparison to CMIP5 models. *Biogeosciences* 17, 4173–4222 (2020).
35. S Piao, P Friedlingstein, P Ciais, N Viovy, J Demarty, Growing season extension and its impact on terrestrial carbon cycle in the Northern Hemisphere over the past 2 decades. *Glob. Biogeochem. Cycles* 21, 1–11 (2007).
36. GJ Nabuurs, MJ Schelhaas, GfMJ Mohren, CB Field, Temporal evolution of the European forest sector carbon sink from 1950 to 1999. *Glob. Chang. Biol.* 9, 152–160 (2003).
37. RS Padrón, L Gudmundsson, L Liu, V Humphrey, SI Seneviratne, Drivers of intermodel uncertainty in land carbon sink projections. *Biogeosciences* 19, 5435–5448 (2022).
38. M Fernández-Martínez, et al., Diagnosing destabilization risk in global land carbon sinks. *Nature* 615, 848–853 (2023).

869	39. I Melnikova, et al., Carbon Cycle Response to Temperature Overshoot Beyond 2 °C: An Analysis of CMIP6 Models. <i>Earth's Futur.</i> 9, 1–19 (2021).	931
870		932
871	40. Q Hu, et al., Intercomparison of global terrestrial carbon fluxes estimated by MODIS and Earth system models. <i>Sci. Total. Environ.</i> 810, 152231 (2022).	933
872		934
873	41. S Zechlau, M Schlund, PM Cox, P Friedlingstein, V Eyring, Do Emergent Constraints on Carbon Cycle Feedbacks Hold in CMIP6? <i>J. Geophys. Res. Biogeosciences</i> 127 (2022).	935
874		936
875	42. BK Gier, et al., Representation of the terrestrial carbon cycle in CMIP6. <i>Biogeosciences</i> 21, 5321–5360 (2024).	937
876		938
877	43. CA Sierra, SE Crow, M Heimann, H Metzler, ED Schulze, The climate benefit of carbon sequestration. <i>Biogeosciences</i> 18, 1029–1048 (2021).	939
878		940
879	44. SE Crow, CA Sierra, The climate benefit of sequestration in soils for warming mitigation. <i>Biogeochemistry</i> 161, 71–84 (2022).	941
880	45. VK Arora, et al., Carbon-concentration and carbon-climate feedbacks in CMIP5 earth system models. <i>J. Clim.</i> 26, 5289–5314 (2013).	942
881	46. MD Hurteau, MP North, GW Koch, BA Hungate, Managing for disturbance stabilizes forest carbon. <i>Proc. Natl. Acad. Sci. United States Am.</i> 116, 10193–10195 (2019).	943
882		944
883	47. H Li, T Ilyina, T Loughran, A Spring, J Pongratz, Reconstructions and predictions of the global carbon budget with an emission-driven Earth system model. <i>Earth Syst. Dyn.</i> 14, 101–119 (2023).	945
884		946
885	48. N Wei, et al., Evolution of Uncertainty in Terrestrial Carbon Storage in Earth System Models from CMIP5 to CMIP6. <i>J. Clim.</i> 35, 5483–5499 (2022).	947
886		948
887		949
888	49. M O'Sullivan, et al., Aerosol-light interactions reduce the carbon budget imbalance. <i>Environ. Res. Lett.</i> 16 (2021).	950
889		951
890	50. H Yang, et al., Global increase in biomass carbon stock dominated by growth of northern young forests over past decade. <i>Nat. Geosci.</i> 16, 886–892 (2023).	952
891		953
892	51. CL Quéré, et al., Drivers of declining co2 emissions in 18 developed economies. <i>Nat. Clim. Chang.</i> 9, 213–217 (2019).	954
893	52. H Qiu, D Hao, Y Zeng, X Zhang, M Chen, Global and northern-high-latitude net ecosystem production in the 21st century from CMIP6 experiments. <i>Earth Syst. Dyn.</i> 14, 1–16 (2023).	955
894		956
895	53. AD Friend, et al., Carbon residence time dominates uncertainty in terrestrial vegetation responses to future climate and atmospheric CO2. <i>Proc. Natl. Acad. Sci. United States Am.</i> 111, 3280–3285 (2014).	957
896		958
897	54. E Muñoz, I Chanca, CA Sierra, Increased atmospheric CO2 and the transit time of carbon in terrestrial ecosystems. <i>Glob. Chang. Biol.</i> 29, 6441–6452 (2023).	959
898		960
899	55. E Muñoz, et al., On the importance of time in carbon sequestration in soils and climate change mitigation. <i>Glob. Chang. Biol.</i> 30, 10–12 (2024).	961
900		962
901		963
902	56. Z Shi, et al., Global-Scale Convergence Obscures Inconsistencies in Soil Carbon Change Predicted by Earth System Models. <i>AGU Adv.</i> 5, 1–11 (2024).	964
903		965
904	57. F Ge, S Zhu, H Luo, X Zhi, H Wang, Future changes in precipitation extremes over southeast asia: Insights from cmip6 multi-model ensemble. <i>Environ. Res. Lett.</i> 16 (2021).	966
905		967
906	58. V Eyring, et al., Overview of the coupled model intercomparison project phase 6 (cmip6) experimental design and organization. <i>Geosci. Model. Dev.</i> 9, 1937–1958 (2016).	968
907		969
908	59. APK Argles, JR Moore, PM Cox, Dynamic Global Vegetation Models: Searching for the balance between demographic process representation and computational tractability. <i>PLOS Clim.</i> 1, e0000068 (2022).	970
909		971
910	60. E Weng, et al., Modeling demographic-driven vegetation dynamics and ecosystem biogeochemical cycling in NASA GISS's Earth system model (ModelE-BiomeE v.1.0). <i>Geosci. Model. Dev.</i> 15, 8153–8180 (2022).	972
911		973
912	61. M Kondo, et al., State of science in carbon budget assessments for temperate forests and grasslands in Balancing Greenhouse Gas Budgets. (Elsevier), pp. 237–270 (2022).	974
913		975
914	62. C Rödenbeck, Estimating CO2 sources and sinks from atmospheric mixing ratio measurements using a global inversion of atmospheric transport, (Max-Planck Institute for Biogeochemistry), Technical Report 6 (2005).	976
915		977
916		978
917		979
918	63. F Chevallier, On the parallelization of atmospheric inversions of CO2surface fluxes within a variational framework. <i>Geosci. Model. Dev.</i> 6, 783–790 (2013).	980
919		981
920	64. J Zhuang, et al., pangeo-data/xESMF: v0.8 (2023).	982
921	65. KP Burnham, DR Anderson, eds., Model selection and multimodel inference. A practical information-theoretic approach. (Springer New York, New York, NY), 2 edition, p. 488 (2002).	983
922		984
923		985
924		986
925		987
926		988
927		989
928		990
929		991
930		992